

Milan Barot

7milan16@gmail.com | (917) 497-6979 | DMV, VA | [linkedin.com/in/7mb16](https://www.linkedin.com/in/7mb16) | <https://70milan.github.io/>

SUMMARY

Data Engineer with 6 years of experience specializing in **healthcare data engineering, payer/TPA systems, ETL, and legacy-to-modern platform migration**. Experienced with **HealthRules Payer (HRP), claims, eligibility, accumulator, provider, and financial data**, building and optimizing SQL-based data pipelines and reporting solutions across on-premises and cloud environments. Proven track record of translating complex healthcare business requirements into scalable data solutions while ensuring data accuracy, reliability, and production readiness. Currently expanding into **modern data engineering with Apache Spark, Databricks, and lakehouse architectures**, with hands-on experience building end-to-end data platforms using **Medallion architecture, Delta Lake, and AI-powered analytics**.

TECHNICAL SKILLS

Data Engineering	SQL, Python, PySpark, Apache Spark, SSIS, Azure Data Factory, ETL
Cloud & Lakehouse	Azure, Databricks, Delta Lake, Medallion Architecture
Databases/Data Warehouse	SQL Server, Azure SQL, PostgreSQL, DuckDB
BI & Reporting	Power BI, SSRS
Data Integration	EDI 834/837, REST APIs, FTP, ActiveBatch, SQL Server Agent
Workflow & DevOps	Airflow, Dagster, Docker, Git, Jenkins, Jira
Application & AI	FastAPI, React, Electron, ChromaDB, OpenAI API, Groq API

WORK EXPERIENCE

AmeriBen (UST Global | Implementation partner), remote (Health insurance)

SQL Server Developer/ Data Engineer January 2026 – Current

- Contributed to the migration of healthcare reporting from a legacy **GBAS** platform to an **HRP-based ODS** architecture, supporting the onboarding of self-funded employer groups and transition of **financial, utilization, claims, census, and clinical reporting** to the new platform.
- Developed and maintained **complex SQL Server reporting logic, stored procedures, merge scripts, and ETL workflows** supporting reports for **large claims, financial summaries, contract savings, utilization, lag analysis, and level-of-coverage census** across legacy and modernized data sources.
- Built **consolidated reporting datasets spanning both GBAS and ODS** during the platform transition, reconciling fundamentally different schemas and business structures to ensure consistent reporting for clients operating across legacy and HRP-derived data.
- Investigated data discrepancies through the **HPS → ODS transformation pipeline**, analyzing source HRP data, mappings, ETL logic, and downstream reporting objects; implemented corrective changes to ensure accurate representation of migrated healthcare data.
- Translated complex **payer/TPA business requirements** into production-grade SQL and reporting solutions, resolving ETL issues, SQL performance bottlenecks, cross-database dependencies, and data-quality issues across development, testing, and production environments.
- Collaborated with **business stakeholders and HRP teams** to understand healthcare reporting requirements, validate data mappings and business rules, and deliver technical solutions aligned with project requirements and migration milestones.

Tech Stack: SQL Server 2022, Azure SQL, SSRS, SSIS, ETL, HealthRules Payer (HRP)

Blue Cross Blue Shield Arizona (Mastek | Implementation partner), Phoenix, AZ (Health insurance)

Data Engineer Sept 2022 – Oct 2025

- Designed and implemented end-to-end **ETL pipelines** using **SSIS** and **SQL Server** to process **eligibility, claims, accumulator, and value-based care (VBC)** files, ensuring compliance with business requirements and vendor-specific standards. Collaborated with vendors such as **OptumRx, HealthEquity, Equality Health**, and others to deliver high-quality data solutions.
- Oversaw a small team of offshore resources by recruiting and onboarding team members as required for project execution. Ensured tasks were completed on time by maintaining close collaboration with both onshore and offshore teams
- Utilized **Azure Data Factory (ADF)** to design and orchestrate scalable **data integration pipelines** for migrating legacy member and claims data to **HealthRules Payer (HRP)** on **Azure Cloud**.

- Leveraged **Azure SQL Database** for staging, transformation, and persistent storage of high-volume healthcare data. Configured **ADF linked services, pipelines, and triggers** for automated, scheduled data ingestion from on-premises SQL Server to Azure.
- **Rebuilt critical ETL pipelines** using **SSIS** and **SQL Server** after the **CHC system hack (2024)**, restoring high-priority data workflows under tight deadlines with minimal downtime.
- Developed **complex data extracts** that combined on-prem warehouse data with select **cloud-based sources**, ensuring consistency, accuracy, and seamless integration across hybrid environments reducing execution time by 25 % on an average across the board.
- Collaborated closely with **Deloitte** on the HealthEquity project to develop **eligibility and claims files** and automated it end to end. Responsibilities included gathering business requirements, creating technical documentation, optimizing ETL pipelines, conducting performance testing, and ensuring smooth deployment into production environments.
- Automated data workflows by integrating **SQL Server Agent jobs** with **ActiveBatch**, setting up FTP connectivity with strict naming conventions and cadence-driven schedules for file delivery to vendors.
- Developed a consolidated **EDI framework for 834 and 837 enrollments, claims and dental**, using three core SSIS packages to automate the generation and delivery of over **35 distinct file types** to **CMS**. This architecture ensured secure data exchange and audit readiness while significantly reducing the manual maintenance and operational overhead previously required.
- Provided active production support for mission-critical **CMS data exchanges**, conducting over 25 successful production deployments and managing emergency releases to maintain 100% compliance with federal reporting deadlines.
- Led the rapid resolution of critical production failures, ensuring zero data loss during **high-volume enrollment** periods where direct connectivity with **CMS** was required
- Partnered with business stakeholders to gather requirements for new projects, conducting detailed assessments of complexity, technical effort, development timelines, and warranty periods. Worked with Mastek to draft **SOWs** for approval by Blue Cross Blue Shield, ensuring alignment with business objectives.
- Participated in Agile sprint meetings to track progress on deliverables, resolve blockers, and ensure timely completion of project milestones across multiple initiatives.

Tech Stack: Azure Data Factory (ADF), Azure SQL Database, C#, SQL Server, SSIS, SSRS, ActiveBatch, SQL Server Agent, FTP, EDI (834, 837), HealthRules Payer (HRP), Agile +4

CBRE, Dallas, TX (Commercial real estate services and investments)

Data Engineer Feb 2021 - Sept 2022

- Played a key role on the **Data Engineering & Reporting team** during the COVID-19 pandemic, delivering real-time data pipelines and analytics that supported CBRE's **Advisory business unit** across global commercial real estate portfolios.
- Designed, developed, and maintained robust **ETL pipelines** using **Python, SQL, and PostgreSQL** via CBRE's internal ETL framework (**Lymbic**). Translated complex business logic from real estate analysts into SQL/Python transformations for property usage, tenant risk, and leasing performance.
- Ingested diverse external data sources such as **Google Mobility Reports** and **BREAM energy efficiency assessments**, transforming them into structured datasets in **Parquet** and **CSV** formats. These outputs fed into **PowerBI dashboards** used by leadership to assess region-wise COVID impacts on commercial spaces.
- Applied data profiling and transformation using **Pandas** and **PySpark** for efficient in-memory processing of large CSV/Parquet files, enabling faster QA cycles and stakeholder feedback loops.
- Supported **Databricks** orchestration workflows — monitoring job execution, rerunning failed tasks, and assisting engineers with job recovery — ensuring daily and weekly batch pipelines remained consistent and reliable during volatile COVID data conditions.
- Worked in an **Agile delivery environment**, actively collaborating with analysts, product managers, and backend engineers in weekly sprint meetings to prioritize features, track bugs, and refine pipeline outputs based on stakeholder needs.
- Enabled high-impact real estate decisions by producing clean, trusted data for dashboards that revealed critical trends such as **building underutilization, shifting tenant behaviors, and geographic leasing risk**. These insights directly influenced CBRE's portfolio restructuring strategies and client advisory recommendations during the pandemic.

Tech Stack: Python, SQL, PostgreSQL, Power BI, Airflow, Pandas, Parquet, Agile

AmNet Services Inc., Fairfield, NJ (Telecommunications)

ETL/SQL Server Developer Oct 2019 - Aug 2020

- Developed SSIS packages to automate extraction and ingestion of network test data from Excel, flat files, and MS Access sources into a SQL Server staging environment.
- Designed and implemented a dimensional data warehouse model using **ERwin Data Modeler** to support analytical queries on network KPIs.

- Wrote complex **T-SQL stored procedures** for **data cleansing, aggregation, and validation** to ensure accuracy and consistency of telecom metrics.
- Created and maintained **SSRS reports** and **PowerBI dashboards** to visualize signal strength, call drop rates, and E911 test call success metrics.
- Automated **SQL Server Agent** jobs to schedule SSIS package executions and report generation, ensuring daily data processing without manual intervention.
- Collaborated with radio engineers, business analysts, and compliance teams in Agile sprints to gather requirements and iteratively improve data solutions.
- Conducted rigorous testing of ETL workflows and reports to guarantee data quality and system performance at scale.

Tech Stack: SSIS/SSRS, T-SQL, Power-BI, Agile

SELECTED PROJECTS

Kalshi-Prediction-Engine (AI-Powered Intelligence Lake-House) https://github.com/70milan/Kalshi_PredictionProject

- Architected a full-stack prediction market intelligence platform using a **Medallion (Bronze/Silver/Gold)** data lakehouse, processing 6 global news feeds, GDELT event streams, and Kalshi prediction market data through **PySpark** ETL with **Delta Lake ACID** guarantees.
- Engineered a **RAG**-powered inference engine that embeds Silver-layer data into **ChromaDB** (MiniLM-L6-v2), performs cascading time-windowed semantic search, and synthesizes bull/bear/verdict intelligence briefs via **Groq** (Llama 3.3 70B) — including automated "Ghost Pump" detection for unsupported price movements.
- Built a React Human-in-the-Loop trade terminal backed by a **FastAPI** bridge server, surfacing **AI-generated intelligence** briefs with Kelly Criterion bet-sizing and a Safe Mode guard for risk-controlled live trade execution on Kalshi via RSA-signed API calls.
- Deployed a 10-container **Docker** Compose production environment with a **custom ETL orchestrator** (**watermark-based** change detection, lock-file API coordination, once-daily settlement sweeps) running autonomously at \$0/month on local Windows infrastructure.
- Implemented resilient, low-cost ingestion across BBC, CNN, Fox, NYT, NY Post, and The Hindu using RSS discovery, Trafilatura full-text extraction, and **VADER sentiment scoring**, with surgical Kalshi series-first filtering reducing **API** calls by ~98%.

Tech Stack: Python (PySpark/Pandas), GroqAPI DuckDB, ChromaDB, Delta Lake, Kalshi API, GDELT GKG, Sentiment Analysis (VADER/LLM), Docker, React.

RealTime_Context_Engine (AI-Powered Windows Runtime Host) https://github.com/70milan/RealTime_Context_Engine

- Developed a **dual-process desktop application** utilizing an **Electron** frontend and a **Python/FastAPI** backend to provide **real-time** interview guidance.
- Integrated **OpenAI GPT models** via **low-latency streaming (SSE)** to deliver generative feedback with a **<200ms TTFT** (Time To First Token).
- Built a custom **hardware-locked licensing system** using **RSA digital signatures** and **machine-specific HWIDs** to secure distribution.
- Engineered a high-performance **screen capture and audio processing pipeline** for real-time context analysis during live sessions.
- Automated a full **distribution pipeline** using **PyInstaller** and **Electron Builder**, incorporating **code signing** and an **automated update system**.

Tech Stack: Python (FastAPI/Uvicorn), JavaScript (Electron/Node.js), OpenAI API (GPT-4o/5), RSA Cryptography, SSE (Server-Sent Events), PyInstaller, Electron Builder.

EDUCATION

Bachelor of Science in Information Technology in Software Engineering March 2022

Washington University of Science & Technology, Vienna, VA

Bachelors in Applied Science in Computer Information Systems Dec 2019

The City University of New York, New York, New York

CERTIFICATIONS

Databricks Data Engineer Associates Oct 2026

